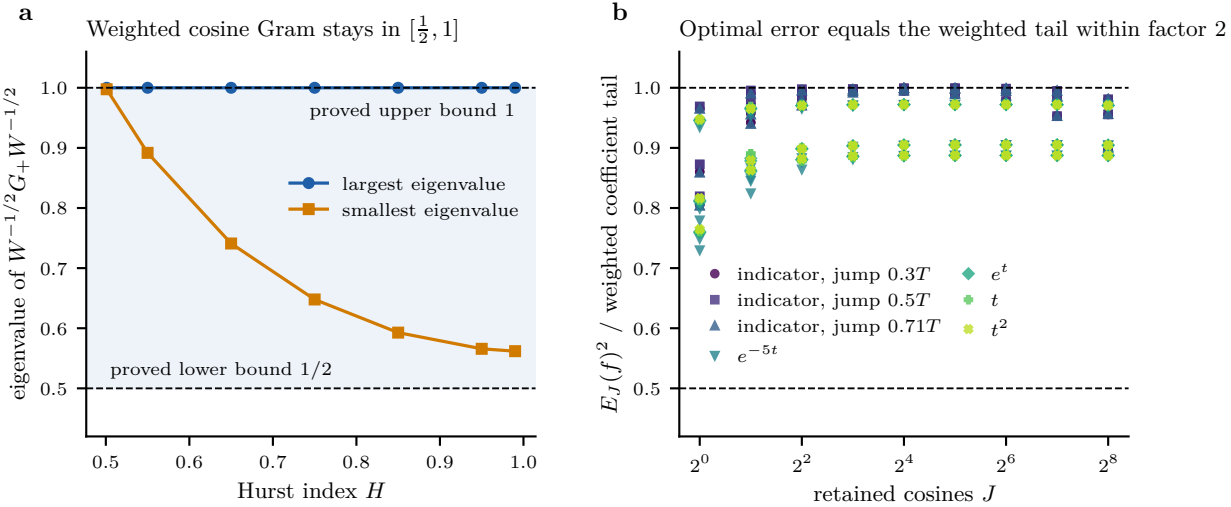

a
Weighted cosine Gram stays in $[\frac{1}{2}, 1]$
proved upper bound 1
largest eigenvalue
smallest eigenvalue
proved lower bound 1/2
eigenvalue of $W^{-1/2} G_+ W^{-1/2}$
1.0
0.9
0.8
0.7
0.6
0.5
0.5
0.6
0.7
0.8
0.9
1.0
Hurst index $H$
b
Optimal error equals the weighted tail within factor 2
$E_J(f)^2$ / weighted coefficient tail
indicator, jump $0.3T$
indicator, jump $0.5T$
indicator, jump $0.71T$
$e^{-5t}$
$e^t$
$t$
$t^2$
1.0
0.9
0.8
0.7
0.6
0.5
$2^0$
$2^2$
$2^4$
$2^6$
$2^8$
retained cosines $J$